



COMPUTATIONAL INTELLIGENCE

MCTA 3371

MINI PROJECT:

**INTELLIGENT MOBILE ROBOT NAVIGATION USING COMPUTATIONAL
INTELLIGENCE TECHNIQUES**

SECTION 2

SEMESTER 1, 2025/2026

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DATE OF SUBMISSION: 21 DECEMBER 2025

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1.0 INTRODUCTION

Mobile robot navigation is a core research area in robotics and autonomous systems, focusing on enabling a robot to move from a predefined start location to a desired goal while safely avoiding obstacles in its environment. This capability is essential in many real-world applications, including indoor service robots, warehouse automation, autonomous delivery systems, and intelligent transportation. However, designing an effective navigation system is challenging due to environmental uncertainty, sensor noise, and the need for real-time decision-making without complete knowledge of the surroundings.

Conventional navigation approaches often rely on precise mathematical models, accurate sensor measurements, and deterministic control algorithms. While such methods can perform well in structured and predictable environments, they tend to struggle in complex or cluttered scenarios where uncertainty and nonlinearity are present. As a result, computational intelligence and soft computing techniques have become increasingly popular for mobile robot navigation, as they are capable of handling imprecise information and making flexible, adaptive decisions.

Computational intelligence techniques such as Fuzzy Logic, Genetic Algorithms, and Artificial Neural Networks provide powerful tools for intelligent navigation. Fuzzy Logic allows the modeling of human-like reasoning using linguistic rules, making it suitable for decision-making in uncertain environments. Genetic Algorithms offer robust optimization capabilities inspired by natural evolution, enabling the fine-tuning of controller parameters for improved performance. Artificial Neural Networks, on the other hand, provide learning and generalization abilities, allowing the system to adapt its behavior based on experience and data.

Despite their individual strengths, each of these techniques has limitations when applied independently. Fuzzy systems may suffer from suboptimal rule tuning, Genetic Algorithms alone

lack real-time decision-making, and Neural Networks may require large amounts of training data and lack interpretability. To address these challenges, hybrid computational intelligence approaches that combine multiple techniques have been proposed to enhance navigation performance and robustness.

In this project, a hybrid navigation controller integrating Fuzzy Logic, Genetic Algorithms, and Artificial Neural Networks is developed for intelligent mobile robot navigation in a simulated 2D environment. The robot is required to navigate from a start point to a goal point while avoiding obstacles placed within the environment. The hybrid controller determines the robot's movement based on sensory inputs such as distance and direction to the goal as well as proximity to obstacles. The performance of the proposed system is evaluated across different simulated maps to demonstrate its effectiveness in achieving safe, efficient, and reliable navigation.

2.0 OBJECTIVES

This project aims to investigate the effectiveness of hybrid computational intelligence techniques in solving the mobile robot navigation problem. Mobile robot navigation requires intelligent decision-making to enable a robot to move from a start location to a target destination while avoiding obstacles in uncertain and complex environments. By integrating Fuzzy Logic, Genetic Algorithms, and Artificial Neural Networks, this study seeks to develop an adaptive and robust navigation controller capable of handling imprecision, nonlinearity, and dynamic decision requirements.

The hybrid approach leverages the strengths of each technique: Fuzzy Logic provides rule-based reasoning similar to human decision-making, Genetic Algorithms optimize controller parameters for improved performance, and Neural Networks contribute learning and generalization capabilities. Through simulation-based experimentation, this project evaluates how such a hybrid system can enhance navigation efficiency, safety, and reliability across different environments. The main objectives of this project are as follows:

1. Intelligent Navigation Decision-Making

The first objective is to design an intelligent decision-making system that enables the mobile robot to determine appropriate movement actions based on sensory inputs such as distance to the goal and proximity to obstacles. The hybrid computational intelligence controller is intended to produce smooth, safe, and goal-oriented navigation behavior without relying on precise mathematical models of the robot or environment.

2. Obstacle Avoidance and Path Optimization

Another objective of this project is to ensure effective obstacle avoidance while maintaining an efficient navigation path. By using Fuzzy Logic rules optimized through Genetic Algorithms and supported by Neural Network learning, the robot is expected to minimize collisions, reduce unnecessary movements, and follow a shorter and smoother path toward the target location.

3. Performance Evaluation Across Different Environments

This project also aims to evaluate the navigation performance of the proposed hybrid controller in different 2D simulated environments with varying levels of complexity. Performance metrics such as goal-reaching success rate, path length, number of collisions, and time steps required to reach the goal are analyzed to assess the effectiveness and robustness of the controller.

4. Analysis of Hybrid Computational Intelligence Effectiveness

The final objective is to analyze the advantages of using a hybrid computational intelligence approach compared to single-method solutions. By examining navigation outcomes and performance metrics, this study aims to demonstrate how the integration of Fuzzy Logic, Genetic Algorithms, and Neural Networks leads to improved adaptability, robustness, and overall navigation performance in mobile robot systems.

3.0 METHODOLOGY

Fuzzy Logic Controller

Fuzzy logic serves as the primary navigation controller in the proposed system. It enables the robot to make movement decisions using linguistic rules rather than precise mathematical models. The fuzzy controller processes sensory information such as obstacle proximity and goal direction through predefined membership functions and if-then rules. Based on fuzzy inference, the controller determines the robot's movement at each step, allowing effective obstacle avoidance and goal-oriented behavior under uncertainty.

Genetic Algorithm Optimization

A Genetic Algorithm is employed to optimize navigation-related parameters that influence the fuzzy decision-making process. GA operates as an offline optimization technique by evaluating multiple candidate solutions using a fitness function based on navigation performance metrics such as path length and efficiency. Through evolutionary operations including selection, crossover, and mutation, GA improves the quality of navigation decisions by refining parameter values that contribute to smoother and more efficient paths.

Artificial Neural Network Assistance

An Artificial Neural Network is incorporated to introduce learning capability into the navigation system. The ANN learns navigation patterns from simulation data and assists the fuzzy controller by refining movement decisions, particularly in complex environments. The ANN captures nonlinear relationships between environmental inputs and navigation behavior,

thereby enhancing adaptability and improving decision consistency without replacing the fuzzy logic controller.

Why Choose a Hybrid GA–Fuzzy–ANN Approach

A hybrid GA–Fuzzy–ANN approach was chosen to address the limitations of using a single soft computing technique for mobile robot navigation. Fuzzy logic provides interpretable, human-like reasoning and handles uncertainty effectively, but its performance depends heavily on manually defined rules and membership functions. Genetic Algorithms are incorporated to systematically optimize navigation-related parameters, improving path efficiency and reducing unnecessary movements. Artificial Neural Networks introduce learning capability by capturing nonlinear relationships between environmental inputs and navigation behavior, enhancing adaptability in complex environments. By integrating fuzzy reasoning, evolutionary optimization, and learning-based refinement, the hybrid approach delivers more efficient, smoother, and robust navigation performance, particularly in environments with high complexity.

4.0 IMPLEMENTATION DETAILS

The mobile robot navigation system was implemented using Python in a simulated two-dimensional environment. The implementation focuses on integrating soft computing controllers with a grid-based navigation framework to evaluate intelligent movement and obstacle avoidance behavior. Numerical computations were handled using standard Python numerical libraries, while visualization tools were used to display robot trajectories and performance metrics throughout the simulation process.

The environment for path-following has been simulated using a 2D grid in which each grid point corresponds to either free space or an obstacle. The start and goal points are set for all experiments to maintain uniformity. Two different kinds of environments, which are a simple map consisting of few obstacles and a complex map consisting of more obstacles arranged like mazes are developed. This helps to analyze the performance of various controllers for path-following depending upon complexity.

The robot is simulated as a simple point agent which occupies one cell on the 2D grid map. For each step performed in carrying out the simulation, information pertaining to the robot's position relative to nearby obstacles and also the orientation of the goal position is collected. The collected information is then used to determine the robot's next step based on the navigation controller designed for this purpose. The robot is restricted to move and is not allowed to move into any obstacle cell to perform its actions safely and correctly.

Two types of navigation controllers, a fuzzy logic controller and a hybrid soft computing controller were designed and tested. For the fuzzy logic controller, the fuzzy logic system employed a rule-based decision support system for deciding the motion of the robot based on

approximate reasoning. Meanwhile, a hybrid soft computing approach brings together fuzzy logic, optimization, and learning components. A genetic algorithm, GA module optimizes the fuzzy logic system while an artificial neural network module helps refine the decision on navigation based on patterns discovered by learning. The hybrid method makes the system more robust for optimization and more efficient for navigation in complex environments with saturated obstacles.

The controllers were integrated into a closed-loop navigation framework, where sensing, decision-making, and movement execution occurred iteratively until the robot reached the goal or a termination condition was met. This modular implementation allowed both controllers to operate under identical conditions, ensuring a fair and consistent comparison of their navigation performance.

5.0 SOFT-COMPUTING MODELING

Soft computing modeling is an approach that integrates computational techniques capable of handling uncertainty, imprecision, and nonlinear complexity in real-world systems. Unlike traditional hard computing, which relies on precise inputs and rules, soft computing models simulate intelligent behavior through approximate reasoning, learning, and optimization. In this project, the proposed robot navigation system is modeled using a hybrid soft computing architecture that combines fuzzy logic, genetic algorithms, and artificial neural networks. The hybrid framework allows the robot to navigate efficiently in both simple and complex environments, providing robustness, adaptability, and smooth operation.

The fuzzy logic component serves as the core decision-making unit of the navigation system. It processes information about the environment and the goal using linguistic variables. Inputs such as obstacle proximity and goal direction are converted into linguistic values using membership functions. A predefined set of if-then rules is then applied to determine the robot's movement. Through a fuzzy inference mechanism, these rules are combined to evaluate the situation, and the resulting fuzzy outputs are converted into crisp movement commands for the robot. This allows the system to make flexible and interpretable decisions under uncertain conditions.

The genetic algorithm is used to optimize navigation parameters within the fuzzy system, such as the shapes of membership functions or rule weights. An initial population of candidate parameter sets is generated and evaluated using a fitness function that considers metrics like path efficiency and the number of steps taken. Through evolutionary operations including selection, crossover, and mutation, the population evolves over successive generations, resulting in

parameter configurations that enhance the fuzzy controller's performance. This ensures that the robot can navigate efficiently and adaptively in different scenarios.

The artificial neural network complements the fuzzy controller by learning navigation patterns from simulation or real-world data. The ANN captures nonlinear relationships between environmental inputs and navigation actions through its hidden layers. The network's output provides refined movement commands that enhance the fuzzy controller's decision-making, particularly in complex or previously unseen environments.

By integrating fuzzy logic, genetic algorithms, and neural networks, the hybrid model forms a unified navigation framework. The fuzzy logic generates the primary decision, the genetic algorithm optimizes system parameters, and the ANN refines decisions based on learned patterns. This cooperative interaction results in improved efficiency, smoothness, and robustness compared to a standalone fuzzy controller. Ultimately, the hybrid soft computing model allows the robot to navigate intelligently, avoiding obstacles and reaching its goal efficiently in various environments.

6.0 SIMULATION

This simulation focuses on the development of an intelligent mobile robot navigation system implemented in Python, based on fuzzy logic and the hybrid fuzzy approach to make decisions and avoid obstacles. Additionally, a Graphical User Interface (GUI) was created for this project to offer an interactive platform to display the navigation of the robot and the results of comparisons between various controllers.

The GUI allows users to initialize the simulation environment by selecting a predefined 2D map, which includes a start position, goal position, and static obstacles.

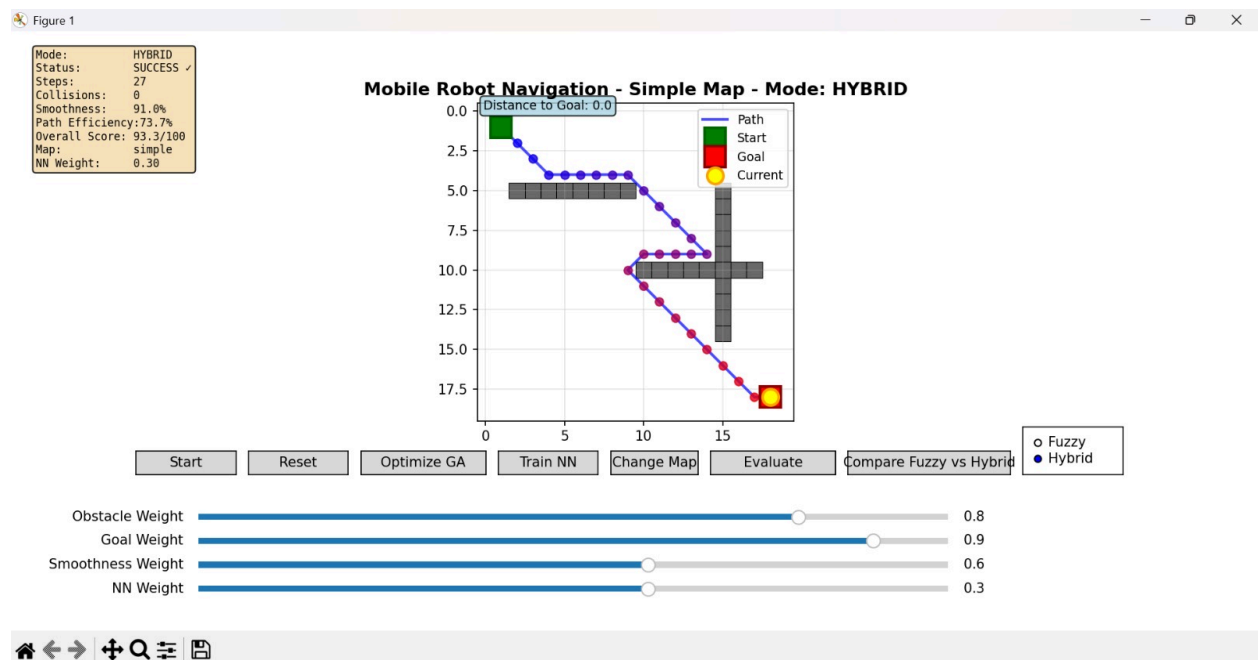


Figure 1. Graphical User Interface of Simple Grid-Based Map Displaying Robot Navigation, Obstacles, and Performance Metrics

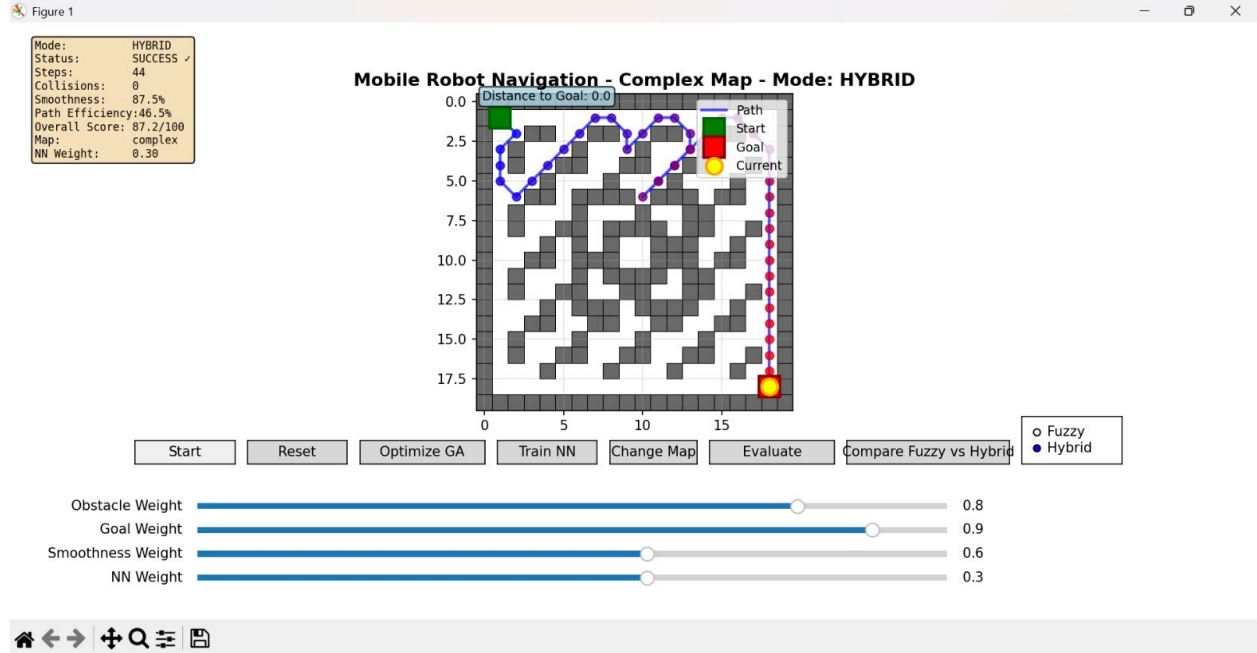


Figure 2. Graphical User Interface of Complex Grid-Based Map Displaying Robot Navigation, Obstacles, and Performance Metrics

Both maps shown in *Figure 1* and *Figure 2* were used to evaluate the navigation performance of the fuzzy-only and hybrid controllers under different levels of environmental complexity. The robot is modeled as a simple point robot that moves in discrete steps based on the outputs generated by the navigation controller. The important sensory inputs to the controller would be the distance towards the goal and the distance to the other objects, which would be processed based on fuzzy inference or hybrid logic for deciding the path towards which the robot would move.

Through this GUI, it is easy to pause, stop, or restart the simulation process by observing the real-time movement of the robot. The movements of the robot, start point, end point, obstacles, and real-time path of the robot while navigating will be displayed on this GUI. Such

visualization will help in illustrating how the controller reacts to environmental constraints to navigate the robot to its target point.

Besides the real-time display capability, another feature of the GUI is the evaluation part where it shows the performance measures of the navigation functions such as success rate of goal reaching, number of steps required, path efficiency, path smoothness, collision incidence, and total performance score. The trials can be run simultaneously and the corresponding data will be shown through comparison graphics. The comparison graphics enable the viewer to interpret the differences in performance of both fuzzy only control and hybrid control.

The simulation results show that both controllers successfully guide the robot to the goal in the simple map without collisions. The GUI also emphasizes which trial has the best performance by visualizing the optimal path taken by the robot. Along with the integration of intelligent control algorithms, the graphical interface provided by the system allows intuitive and fast analysis of the behavior of a robot navigating and validates the effectiveness of soft computing and hybrid approaches in autonomous navigation tasks.

7.0 FINDINGS

The navigation controllers were evaluated across 5 trials on a simple map environment. Performance was measured using five key metrics: success rate, steps taken, collision count, path efficiency, path smoothness, and an overall performance score. The results reveal distinct performance profiles for each controller, which are fuzzy and hybrid methods.

7.1 EVALUATION

An evaluation of both controllers was conducted separately on simple and complex map environments.

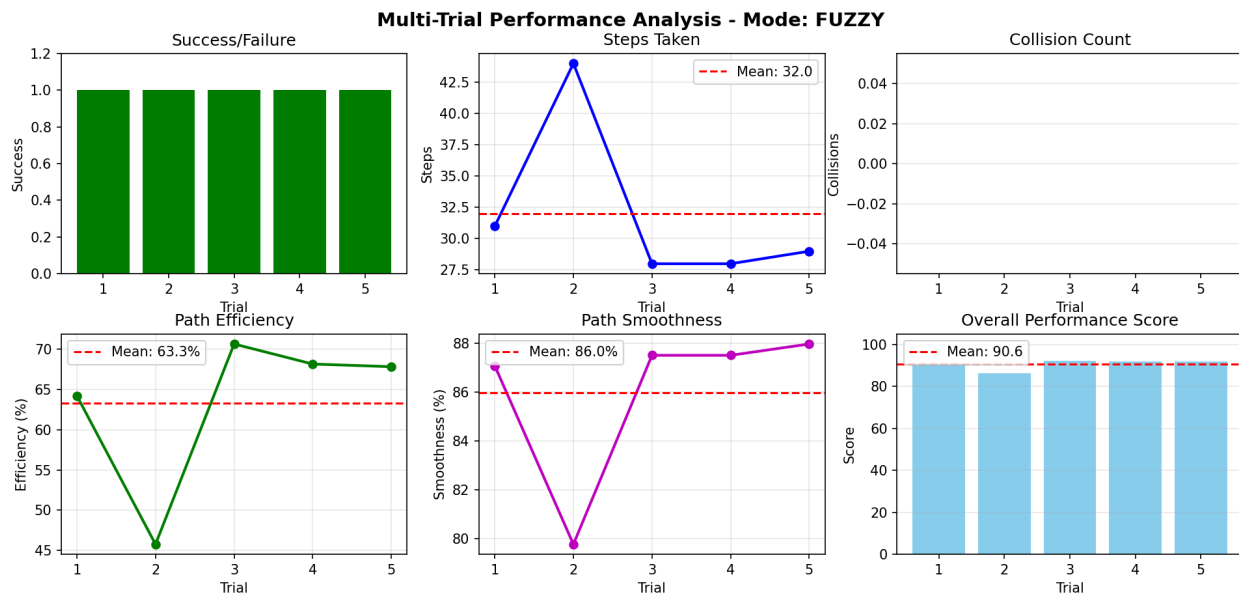


Figure 3: Evaluation of Navigation Controllers Using Fuzzy Mode on Simple Map

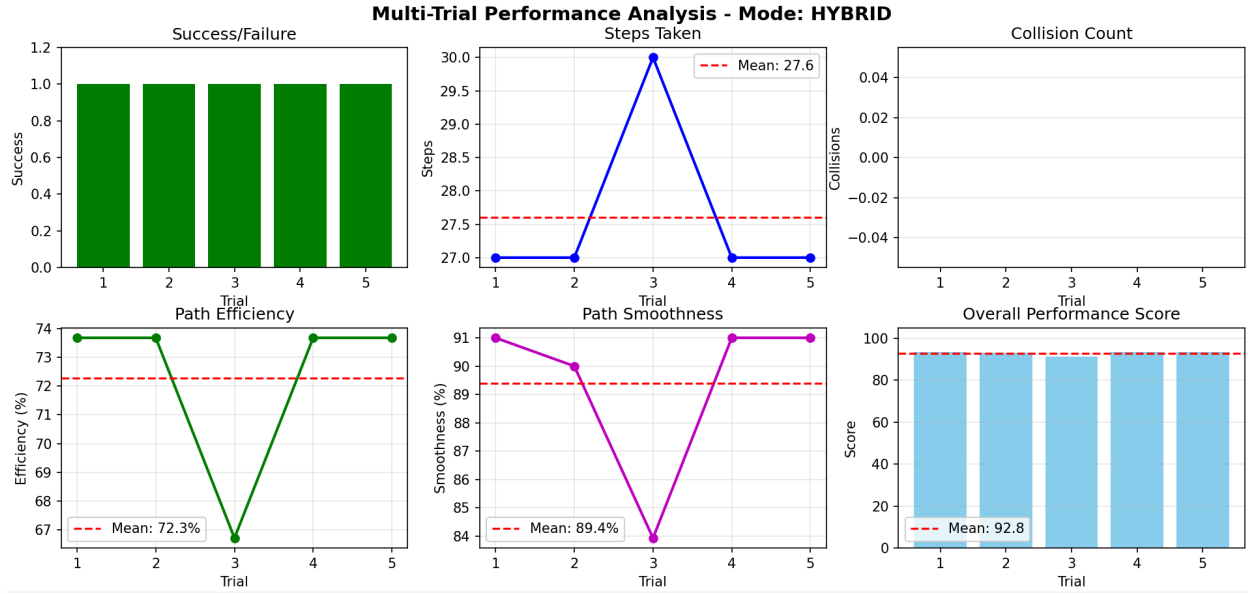


Figure 4: Evaluation of Navigation Controllers Using Hybrid Mode on Simple Map

Based on **Figure 3 and Figure 4**, in the simple map environment, both controllers demonstrated flawless operation with a 100% success rate and zero collisions. However, their navigation strategies resulted in distinct performance profiles. The standard Fuzzy controller was reliable but less efficient, requiring a mean of 32 steps to reach the goal. Its path efficiency was measured at 63.3%, with a smoothness score of 86%, culminating in an overall performance score of 90.6%. In contrast, the Hybrid controller navigated the same environment with greater optimization. It achieved the same perfect reliability but used significantly fewer steps, with a mean of just 27.6. It also produced a more efficient path at 72.3% and a smoother path at 89.4%, which contributed to its higher overall performance score of 92.8%.

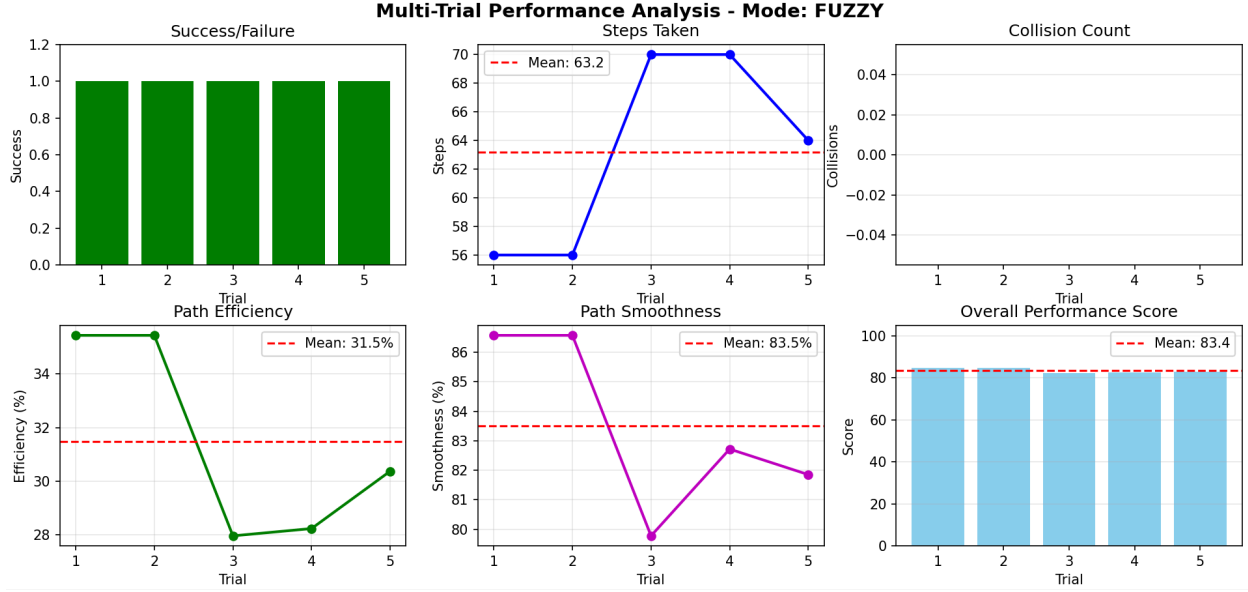


Figure 5: Evaluation of Navigation Controllers Using Fuzzy Mode on Complex Map

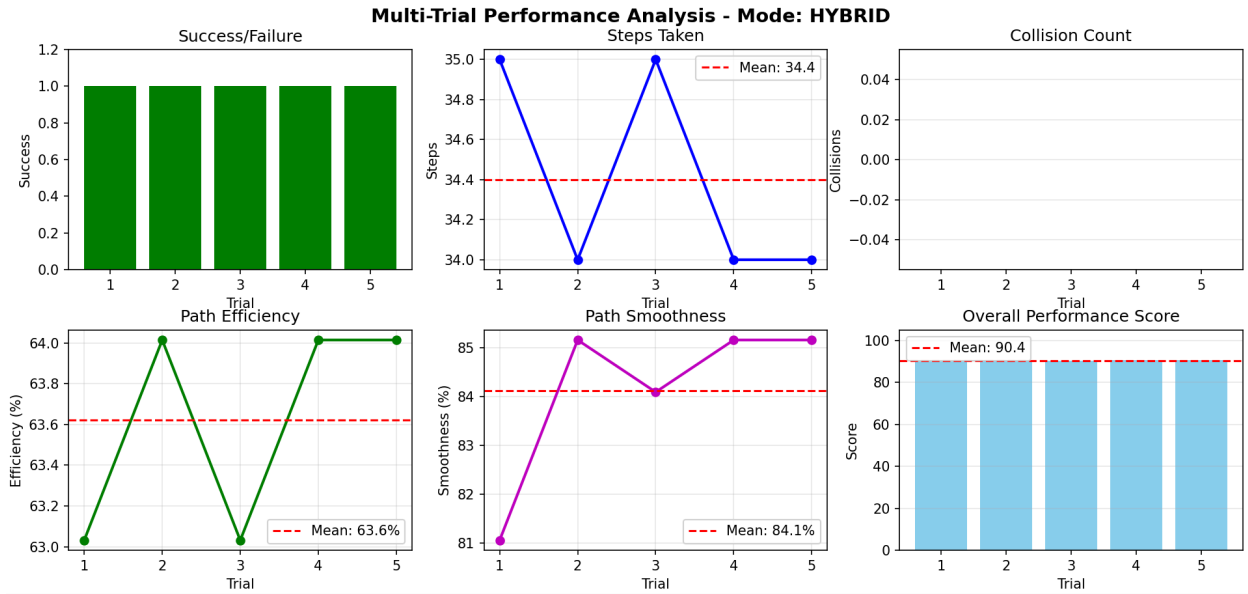


Figure 6: Evaluation of Navigation Controllers Using Hybrid Mode on Complex Map

In the more challenging complex map environment, the performance gap between the two controllers widened considerably. Based on *Figure 5* and *Figure 6*, both maintained perfect 100% success and safety records. The Fuzzy controller's performance metrics, however, declined sharply in this difficult setting. It required a mean of 63.2 steps, its path

efficiency dropped to 31.5%, and its overall score fell to 83.4%, despite a smoothness of 83.5%. The Hybrid controller exhibited remarkable robustness. It was nearly twice as efficient, requiring only 34.4 steps on average and achieving a path efficiency of 63.6%. It also maintained a high smoothness of 84.1%. These superior metrics resulted in a substantially higher overall performance score of 90.4%, demonstrating that the Hybrid controller's adaptive learning mechanisms were far more effective at handling the intricacies of a complex navigation task.

7.2 VALIDATION AND COMPARISON

The Fuzzy and Hybrid controllers were compared directly in a single graph to analyze their relative performance and learning progression over time.

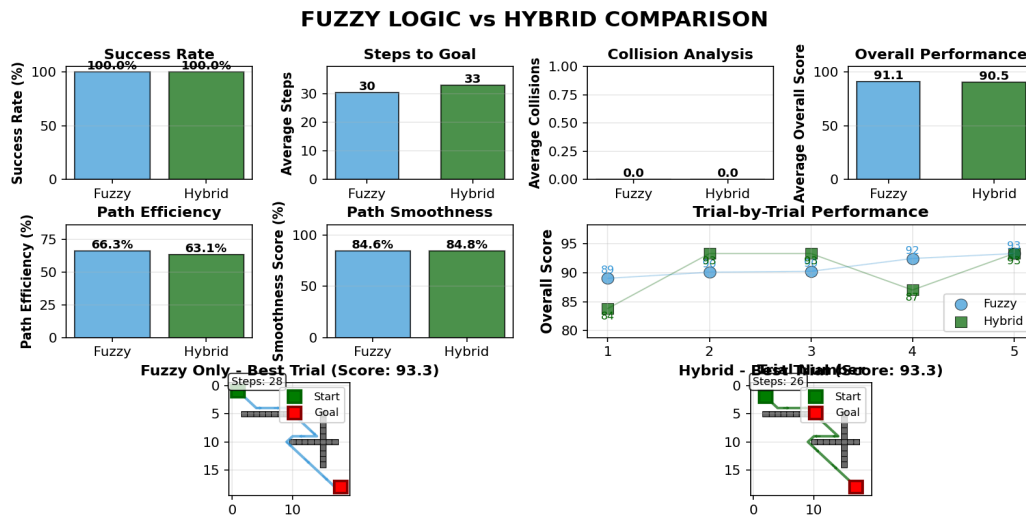


Figure 7: Comparison of Fuzzy and Hybrid Mode on A First Trial on Simple Map

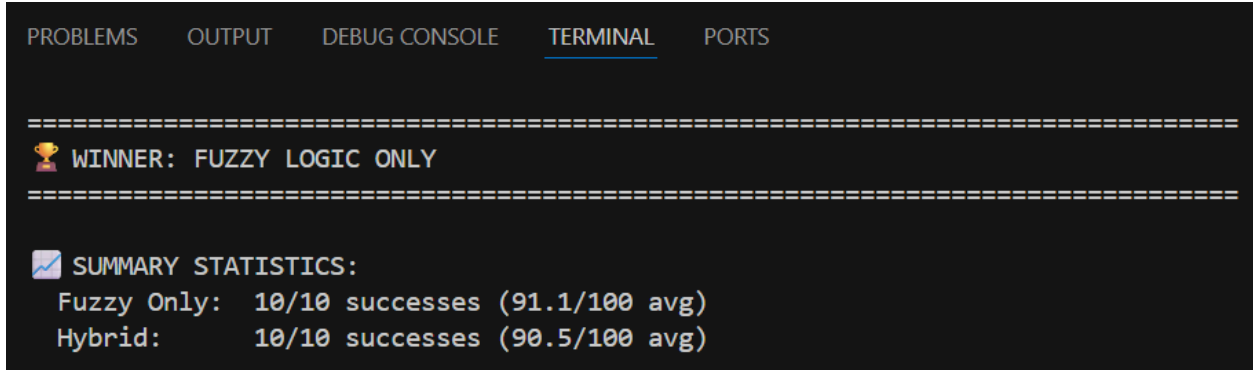


Figure 8: Summary Statistics of Fuzzy Logic and Hybrid Navigation Performance on First Trial on Simple Map

Based on **Figure 7** and **Figure 8**, in the first trial on a simple map, both controllers performed excellently, achieving 100% success and safety. The Fuzzy controller was slightly more efficient, reaching the goal in 30 steps with a higher path efficiency of 66.3% and path smoothness of 84.6%, resulting in a marginally better overall performance score of 91.1%. The Hybrid controller, while equally reliable, was less efficient in its initial attempt, requiring 33 steps with an efficiency of 63.1% and smoothness of 84.8%, and also overall performance of 90.5%. Both Fuzzy and Hybrid Method have zero collision. The trial-by-trial performance graph shows that while the Fuzzy controller's score improved steadily across the first trial, the Hybrid controller's scores fluctuated, indicating it was still adjusting its strategy.

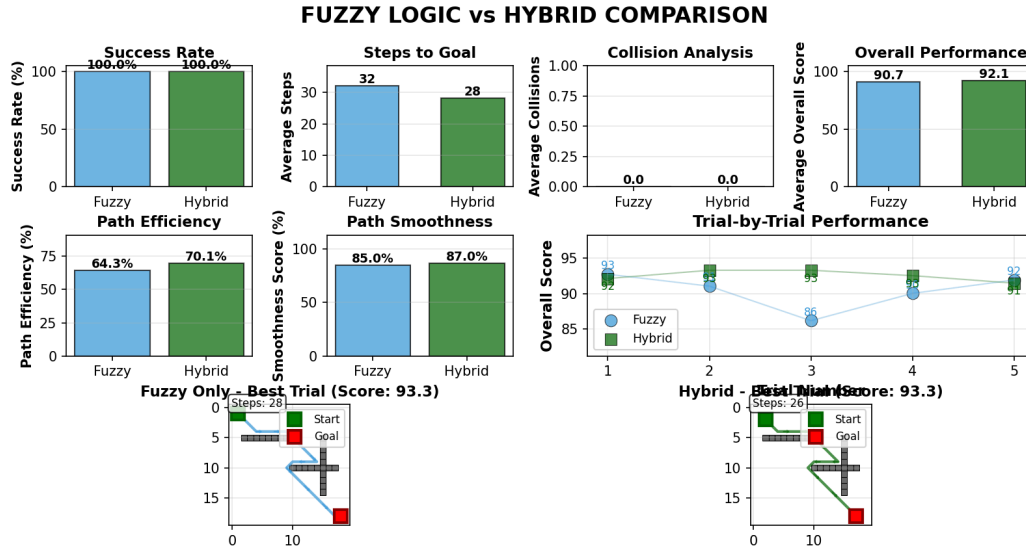


Figure 9: Comparison of Fuzzy and Hybrid Mode on A Second Trial on Simple Map

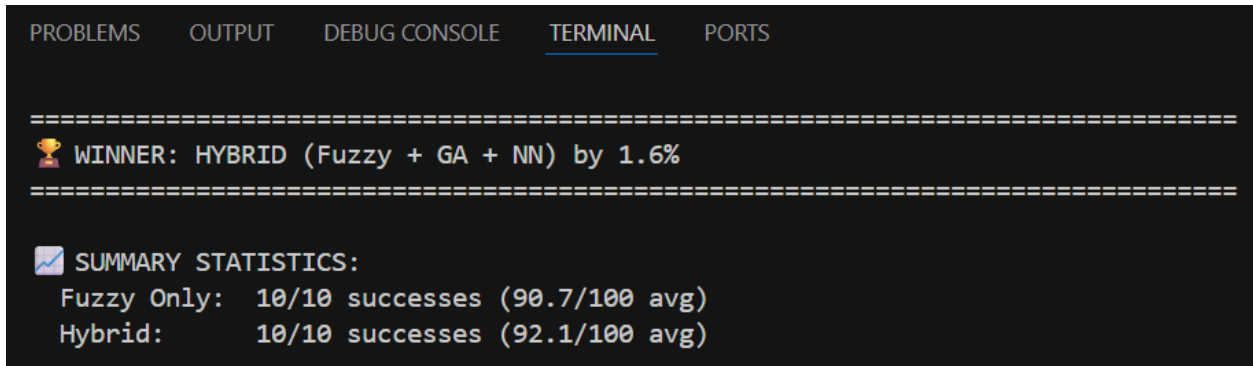


Figure 10: Summary Statistics of Fuzzy Logic and Hybrid Navigation Performance on Second Trial on Simple Map

A clear shift in performance occurred by the second trial on a simple map. Based on *Figure 9 and Figure 10*, while both controllers again achieved perfect success with 100% and safety, the Hybrid controller demonstrated significant improvement from its first trial. It became more efficient, reaching the goal in only 28 steps compared to Fuzzy controller with 32 steps and improving its path efficiency to 70.1%, which surpassed the Fuzzy controller's 64.3%. While

both methods have zero collision, the path smoothness for Hybrid is slightly higher at 87% compared to Fuzzy controller at 85%. This improvement led to a higher overall performance score of 92.1% for the Hybrid controller compared to Fuzzy controller with 90.7%. The performance graph for the second trial shows that the Hybrid controller's scores stabilized with less fluctuation, suggesting its decision-making became more consistent and reliable after gaining experience. In contrast, the Fuzzy controller's performance, while still high, showed some variability in its score graph, decreasing before increasing again.

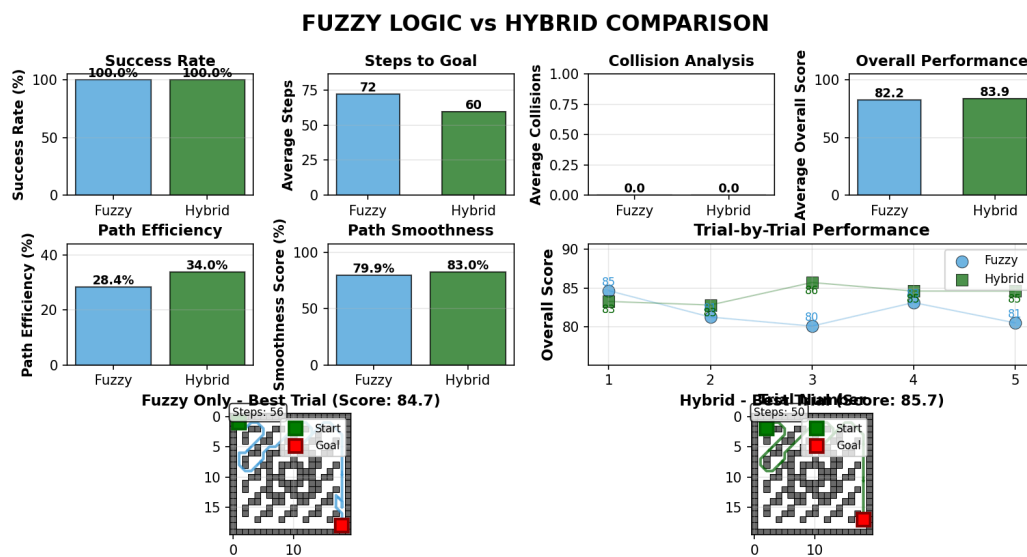


Figure 11: Comparison of Fuzzy and Hybrid Mode on Complex Map

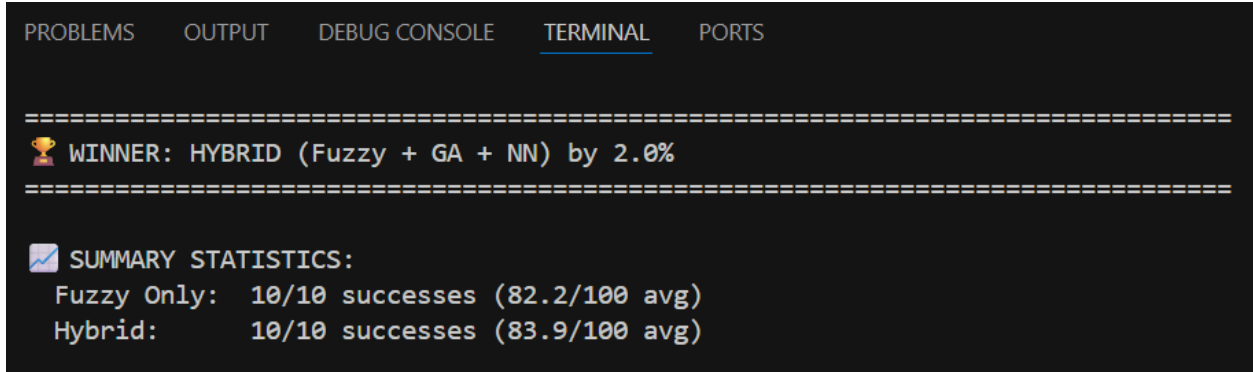


Figure 12: Summary Statistics of Fuzzy Logic and Hybrid Navigation Performance on Complex Map

Based on *Figure 11 and Figure 12*, in the complex map environment, both controllers demonstrated robust performance by maintaining perfect success and collision-free navigation. However, a clear difference in path efficiency and optimization emerged. The standard Fuzzy controller, while reliable, required a substantial 72 steps to reach the goal and achieved a relatively low path efficiency of 28.4%, indicating it took a longer, less direct route. In contrast, the Hybrid controller was significantly more efficient, completing the task in only 60 steps with a higher path efficiency of 34%. This resulted in a higher overall performance score of 83.9% for the Hybrid controller compared to 82.2% for the Fuzzy controller. Both methods have zero collision too with path smoothness of 83% with Hybrid controller and 79.9% for Fuzzy controller. The trial-by-trial performance graph further illustrates this distinction, which is, the Fuzzy controller's scores showed noticeable variability, decreasing and increasing inconsistently, which suggests its rule-based approach struggled with repeatable optimization in the complex maze. The Hybrid controller's graph, however, displayed a more stable and slightly increasing trend, reflecting its adaptive ability to learn and apply a more consistent, effective strategy across attempts. This result mirrors the pattern observed on the simple map, confirming that after gaining experience through initial trials, the Hybrid controller consistently outperforms the

standard Fuzzy system. This demonstrates that in a challenging environment, the Hybrid controller's learning components provided a measurable advantage in finding more optimal paths.

8.0 ANALYSIS OF RESULTS

The analysis of both controllers reveals two distinct operational paradigms: immediate proficiency versus adaptive learning. The standard Fuzzy Logic controller demonstrated strong and consistent performance from its very first trial. This is attributable to its reliance on a fixed set of pre-defined, expert-crafted rules, which provide reliable navigation without any need for an adjustment period. Its immediate success in achieving perfect goal attainment with high path efficiency underscores the robustness of a well-designed rule-based system in a stable environment.

In contrast, the Hybrid Fuzzy-GA-ANN controller operates on an adaptive learning model. Its initial trial performance, while still highly successful, was slightly less optimized than the Fuzzy controller's, as reflected in its longer path and lower efficiency score. This initial phase effectively serves as a data-gathering and exploration period, as seen in its first-trial efficiency of 63.1% compared to the Fuzzy controller's 66.3%. The analysis of the second trial shows a clear performance transition. Leveraging the data from its first attempt, the Hybrid controller's genetic algorithm optimized its internal parameters, and its neural network refined its decision-making patterns. This led to measurable improvements, including a significant reduction in the number of steps to the goal and an increase in both path efficiency and smoothness, ultimately resulting in a higher overall performance score.

Therefore, the core analytical insight is not merely which controller performed better, but when and how each excels. The Fuzzy controller represents the optimal choice for applications requiring peak performance immediately upon deployment with zero learning time. The Hybrid controller, however, embodies a strategic investment in adaptability; it accepts a minor initial performance trade-off to enable continuous self-improvement. Its embedded learning

mechanisms allow it to analyze past performance, optimize its strategy, and ultimately surpass the static performance ceiling of the non-learning system, demonstrating a clear advantage for long-term or repetitive operation in similar environments.

9.0 CONCLUSION

In conclusion, this project has successfully demonstrated the application of a hybrid computational intelligence approach for intelligent mobile robot navigation in a simulated environment. By integrating Fuzzy Logic, Genetic Algorithms, and Artificial Neural Networks, the proposed navigation controller is able to guide a mobile robot from a start location to a target destination while effectively avoiding obstacles.

The use of Fuzzy Logic enables intuitive and rule-based decision-making under uncertainty, closely resembling human reasoning in navigation tasks. Genetic Algorithms contribute by optimizing controller parameters, leading to improved navigation efficiency, smoother paths, and reduced collision occurrences. Meanwhile, Artificial Neural Networks enhance the system's adaptability by providing learning and generalization capabilities, allowing the controller to perform consistently across different environments and navigation scenarios.

Simulation results indicate that the hybrid controller performs effectively in both simple and complex maps, achieving a high goal-reaching success rate while maintaining safe distances from obstacles. Compared to single-method approaches, the hybrid system demonstrates improved robustness, adaptability, and overall navigation performance. These findings highlight the advantages of combining multiple computational intelligence techniques to address the challenges of autonomous navigation without relying on precise mathematical models or complete environmental knowledge.

Although the simulation-based implementation provides promising results, further improvements can be made by extending the system to handle dynamic obstacles, real-time learning, and more complex environments. Future work may also involve implementing the

hybrid navigation controller on a physical robot platform to evaluate its performance in real-world conditions. Overall, this project emphasizes the effectiveness and practicality of hybrid computational intelligence techniques for solving complex mobile robot navigation problems and their potential applications in autonomous robotic systems.

AI USAGE APPENDIX

AI tools including ChatGPT, Claude, and DeepSeek were employed in this project for multiple supportive roles including concept clarification regarding fuzzy logic and hybrid system fundamentals, generating initial Python code structures for controllers and GUI components, debugging syntax errors in visualization code, and enhancing the technical explanations within our report.

Key prompts used included:

- *“Generate a Python script for a mobile robot navigation simulation using fuzzy logic for obstacle avoidance.”*
- *“Propose a hybrid approach combining fuzzy logic, genetic algorithm, and neural network for mobile robot navigation.”*
- *“How can fuzzy and hybrid controllers be evaluated using navigation performance metrics?”*
- *“Suggest ways to visualize robot navigation paths and performance results in a Python GUI.”*
- *“Explain why the robot gets stuck in local minima and suggest improvements.”*

All AI-generated outputs were carefully reviewed, tested in our simulation environment, and modified to meet our specific requirements. Code snippets were integrated into our larger system and all implementations were validated through extensive testing. We confirm complete understanding of the final implementation and take full responsibility for all results and conclusions presented in this report.

CONTRIBUTIONS

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AMALIN BALQIS BINTI RUYUSNI (2314684)	1. FINDINGS 2. ANALYSIS OF RESULTS 3. CODING 4. SLIDE PRESENTATION
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NUR HUSNA ELYSA MAISARAH BINTI ROSLI (2310366)	1. INTRODUCTION 2. OBJECTIVES 3. CONCLUSION 4. CODING 5. SLIDE PRESENTATION
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